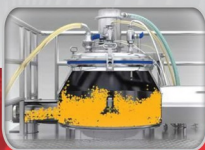




Sameto AG Drive Pvt. Ltd.



HIGH SHEAR MIXING GRANULATOR (RMG)



High Shear Mixing Granulator

application

- Chemiplant's RAPID MIXER GRANULATOR Machine is used for homogeneous mixing of granulation powder used for production of tablets. It is also used for process of wet mixing & fast dry, humidifying of the powder in Chemical, Cosmetic, Food, Plastic, Pharmaceutical, General Mixing Industries, etc.
- Rapid Mixer Granulator series is characterized by compact construction, modern ergonomics design and with all the user & maintenance friendly attributes which are essential for a wide range of application.
- A high-shear mixer can be used to create emulsions, suspensions, lyosols (gas dispersed in liquid) & granular products used in the food, chemical, adhesives, cosmetic, pharmaceutical, and plastics industries for emulsification, homogenization, particle size reduction, & dispersion.



description

The Rapid Mixer Granulator performs dry mixing & wet granulating by basic design of the special three arm of the mixing impeller and sequences of the mixing process in the cylindrical mixing drum with rounded connection to the base plate, achieve another special effect.

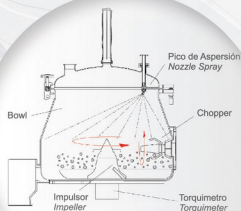
Dry mixing is the first step of the process. Once the products are mixed by bottom drive rotating tool (impeller), a binder is added into the product.

The binder can be sprayed in as part of a binding solution or it can be added in dry form. After this Mixing process, spraying process occurs through one or more spray nozzles and by mixing action of the main mixing tool (impeller) and the chopper blade it is quickly dispersed into the product.



Then Chopper cuts the wet lumps of the material into small sized parts those are mixed equally by the impeller and its speed is responsible for the size of the granules. The dense granules of uniform particle size are produced when the impeller and chopper work together during the wet massing phase, once all solution is in the granulator.

The formulated granules are then discharged from a port in the bowl's side very easily and automatically. Once wet granules come out of the granulator they are generally milled.



Impeller Blade Design

Our impellers are designed in such a way that small blade lift the material and full length blades push the material to mix well. To ensure good circulation of material even at lower speed, the blades with enlarged tips are positioned tangentially to the central hub.



Chopper

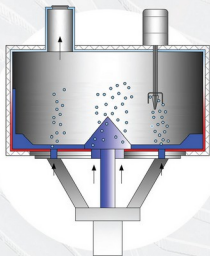
The size of granule is decided by the speed of Choppers. The binder solution dispersion and product movement is improved by multi blade design at slow speeds. Thick products can also be mixed and granulated without any problem. For the precise control of the particle size range and optimal densification of the granules, the cutting edges of chopper are positioned in the zone with circumferential speed.



Wet Milling

Wet milling is done by positioning a particle size reduction milling machine at the discharge port of the high shear granulator. The mill is equipped with a screen specially designed for wet milling with larger openings than typically used for final particle sizing. To ensure the entrapment of any possible ferrous articles resulted during the wet granulation process, a magnetic screen is fixed at the discharge of the wet mill.





Air Purged Seals

Air Purged Seals are specially designed as per CGMP grade to maintain an envelope of air around the entire. The powder transferred into the RMG bowl through a pick n place system where the RMG bowl need not be vacuumized. The other option is where we can vacuumize the bowl by imparting mechanical seals and thereby effecting the powder transfer. The rate of powder transfer would be between 500kg/hr to 1000kg/hr

Clean In Place (CIP)

The RMG is provided with a CIP skid comprising the following:-

- Hot air drying
- Raw water cold wash from inside
- Hot water wash from inside
- Purified water rinse from inside
- Hot water wash with detergent dosing inside

CIP manifold in built into the system washes all expose surfaces including main chamber, shaft seal, chopper blade without external water hoses. The system is pressurized by insitu centrifugal pump. The estimated cycle time for the above sequence of operations with drying is approx.. 15-20 minutes (for largest equipment) A hand-washing gun is provided for cleaning services and certain areas on the equipment, which needs special attention. Also the hand wash gun can be used for washing of equipment from outside, if necessary.



PLC Based Control

PLC based panels are provided to include:-

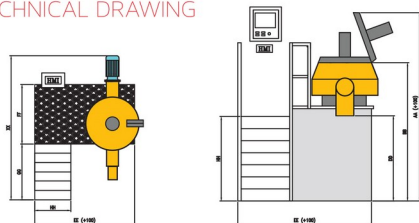
- Programme facility with memory to automatically control liquid addition and granulation end point.
- Programmes for automatic cleaning of the machine.
- Programmes for automatic on/off at preset timing for main impeller and chopper.

Highlight Features

- Air purge sealing for main shaft and granulator shaft.
- Homogeneous binder distribution
- Impeller lifting to a height of 150 mm from bowl bottom for easy cleaning.
- Lifting, swiveling and tilting of lid assembly by 90° for easy cleaning of lid internals.
- Modular design for chopper blades for optimum results.
- Spray nozzle and peristaltic pump for binder granulating liquid addition.
- Jacket for cold /hot water circulation.
- Integrated cleaning system for WIP & CIP.
- PLC based control with recipe administration and visualization.
- Suitable for container and vacuum transfer product infeed.
- Models available from 25 Ltrs. To 1250 Ltrs. capacity.
- Short batch time & reduced cleaning time.
- The seal housing and drive shaft may be flushed with cleaning water, which is then drained away from the machine through built in drain tubes.
- GMP/Flame proof/ SS 316 construction on demand.
- Seven different layouts to suit machine placement as per the room layout.
- Enclosed moving parts. Maximum care has been taken to ensure machine safety.
- Can be used in the fields of Pharmaceuticals/Bulk Drugs and processing of special food stuffs.
- Auto/Manual Controls.



TECHNICAL DRAWING



TECHNICAL SPECIFICATIONS

Dimensions	AA	BB	CC	DD	EE	FF	GG
P 250	2435	1625	800	900	1800	1000	645
P 300	2515	1700	800	900	1800	1000	645
P 400	2775	1940	1200	1095	2200	1365	1045
P 600	3030	1950	1200	1055	2200	1365	1045
P 800	3385	2240	1400	1255	2400	1650	1250
P 1000	3690	2300	1400	1255	2400	1650	1250
P 1250	3755	2350	1400	1265	2600	1850	1250

Technical Data	Bowl volume(l)	Effective bowl volume (l)	Main Motor HP	Tool speed (rpm)	Chopper HP	Chopper speed at 50Hz (rpm)
P 250	235	210	22.5 / 30	5-155	3 / 5.0	1500 / 3000
P 300	285	255	22.7 / 30	5-155	3 / 5.0	1500 / 3000
P 400	430	385	35 / 40	5-125	5 / 7.5	1500 / 3000
P 600	615	550	40 / 50	5-110	5 / 7.5	1500 / 3000
P 800	800	720	50 / 60	5-105	5 / 7.5	1500 / 3000
P 1000	1050	940	63 / 85	5-95	10 / 15.0	1500 / 3000
P 1250	1350	1200	80 / 100	5-90	10 / 15.0	1500 / 3000



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